SOFTWARE REQUIREMENTS SPECIFICATION

QUALITY ASSURANCE ATTENDANCE TRACKER (QAAT)

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Document Control

1.1 Revision History

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1.0.0	2026-04-21	Initial SRS release. Complete system specification for QAAT platform.	MR. LIGHT

1.2 Document Approval

Role	Name	Signature	Date
Author / Architect	MR. LIGHT	________________	2026-04-21
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1.3 Distribution List

Recipient	Organization	Role
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University IT Department	Client Institution	Technical Implementation
Director of Quality Assurance	Client Institution	Requirements Validation
Vice Chancellor Office	Client Institution	Executive Approval

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document provides a complete, formal, and authoritative description of the functional and non-functional requirements for the Quality Assurance Attendance Tracker (QAAT). It has been authored by MR. LIGHT, the System Architect and Product Owner of the QAAT platform.

This document conforms to IEEE Standard 830-1998 (IEEE Recommended Practice for Software Requirements Specifications) and ISO/IEC/IEEE 29148:2018 (Systems and Software Engineering — Life Cycle Processes — Requirements Engineering). It is intended to serve as the binding contractual and technical basis for all design, development, testing, and deployment activities related to the QAAT system.

The primary audience includes university administration stakeholders (Vice Chancellor, Director of Quality Assurance, Quality Assurance Officers), software development teams engaged by MR. LIGHT, IT infrastructure engineers, and third-party integrators connecting existing campus databases with the QAAT platform.

1.2 Scope

QAAT is a multi-tenant, offline-first Software-as-a-Service (SaaS) platform engineered for academic attendance management in university environments. The system eliminates all forms of proxy attendance, administrative bias, and data corruption by combining cryptographic identity verification, physical proximity enforcement via Bluetooth Low Energy (BLE) beacons, decentralised edge computing on a Progressive Web App (PWA), and role-based data access governance.

The QAAT system encompasses the following major sub-systems:

- Cryptographic Identity Issuance Subsystem (QR code generation and email delivery)
- Coordinator PWA Edge Server (Offline-first local session management)
- Physical Proximity Enforcement Layer (BLE Beacon integration)
- Sequential Session Lifecycle Manager (Temporal locking, auto-kill, Lecturer gating)
- Hardware Device Binding Engine (Synthetic device fingerprinting)
- Chunked Resumable Synchronisation Pipeline (Offline-to-cloud data transport)
- Multi-Role Administrative Dashboard Suite (VC, DQA Director, QA Officer)
- Multi-Tenant Policy Engine (Per-institution configuration and threshold management)
- Legacy Database Integration Bridge (Sidecar schema with foreign key linking)

The QAAT system does not replace the university's existing Student Information System (SIS) or Enterprise Resource Planning (ERP) solutions. It operates as an independent but interoperable sidecar platform that extends institutional capability for attendance integrity.

1.3 Definitions, Acronyms, and Abbreviations

Term / Acronym	Definition
QAAT	Quality Assurance Attendance Tracker — the subject system of this SRS.
SRS	Software Requirements Specification — this document.
SaaS	Software-as-a-Service — a software distribution model hosted centrally and accessed via the internet.
PWA	Progressive Web App — a web application that uses service workers to provide native-app capabilities including offline functionality.
BLE	Bluetooth Low Energy — a wireless communication protocol used for proximity beaconing with minimal power consumption.
RSSI	Received Signal Strength Indicator — a measurement of the power level of a received radio signal, used to determine physical proximity.
IEEE	Institute of Electrical and Electronics Engineers.
ISO	International Organisation for Standardisation.
IEC	International Electrotechnical Commission.
RSA	Rivest–Shamir–Adleman — an asymmetric cryptographic algorithm used for QR code signing.
AES-256	Advanced Encryption Standard with 256-bit key — symmetric encryption used for data in transit.
JWT	JSON Web Token — a compact, URL-safe token standard for securely transmitting information between parties.
IndexedDB	A browser-native, high-capacity local database API used by the Coordinator PWA for offline data persistence.
LAN	Local Area Network — a network confined to a limited area, such as a single lecture hall.
HMAC	Hash-based Message Authentication Code — a cryptographic function used to verify QR code integrity.
DQA	Directorate of Quality Assurance — the senior institutional body responsible for academic standards.
VC	Vice Chancellor — the chief academic officer of the university.
QA Officer	Quality Assurance Officer — a field-level administrator responsible for compliance monitoring.
SIS	Student Information System — the university's legacy database containing student records.
Session	A single instance of a lecture or lab, associated with one Course Unit, one Coordinator, one Lecturer, one venue, and one time slot.
Course	An academic programme of study (e.g., Bachelor of Information Technology). Managed by one Coordinator.

Course Unit	A discrete subject within a Course (e.g., Database Systems). Multiple Course Units belong to one Course.
Coordinator	The administrative staff member exclusively assigned to manage QAAT sessions for a given Course.
Warden	A trusted student temporarily delegated by the Coordinator to act as a local session host in the Coordinator's absence.
Atomic Sync	A transactional data synchronisation operation in which all session logs are uploaded as a validated, indivisible batch.
Chunked Resume-Sync	An incremental synchronisation protocol that allows uploads to pause on connection loss and resume from the last confirmed checkpoint.
Ghost Lecture	A fraudulent session record created without a Lecturer being physically present in the classroom.
Proxy Attendance	The fraudulent act of one individual logging attendance on behalf of another.
Edge Server	The Coordinator's PWA device, acting as the local session host and data vault in an offline classroom environment.
Hardware Fingerprint	A synthetic, non-intrusive device identifier generated from immutable hardware and browser characteristics.
RBAC	Role-Based Access Control — a security model in which access rights are assigned according to a user's institutional role.

1.4 References

1. IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications, IEEE Computer Society, 1998.
2. ISO/IEC/IEEE 29148:2018, Systems and Software Engineering — Life Cycle Processes — Requirements Engineering, ISO, 2018.
3. ISO/IEC 25010:2011, Systems and Software Engineering — System and Software Quality Models, ISO, 2011.
4. NIST Special Publication 800-63B, Digital Identity Guidelines — Authentication and Lifecycle Management, NIST, 2017.
5. RFC 7519 — JSON Web Token (JWT), IETF, May 2015.
6. RFC 5246 — The Transport Layer Security (TLS) Protocol Version 1.2, IETF, 2008.
7. Web Bluetooth Community Group Specification, W3C, 2023.
8. Service Workers Level 1, W3C Candidate Recommendation, 2022.
9. QAAT System Architecture Blueprints authored by MR. LIGHT, internal documentation, 2025–2026.

1.5 Overview

The remainder of this document is organised as follows. Section 2 provides a general description of the QAAT product including its context, user characteristics, constraints, and assumptions. Section 3 presents the specific functional and non-functional requirements in full detail. Section 4 documents external interface requirements. Section 5 defines system quality attributes. Section 6 addresses security and compliance requirements. Section 7 covers data requirements. Section 8 defines the deployment and operational environment. Appendices provide supplementary technical information.

2. Overall Description

2.1 Product Perspective

QAAT is a standalone SaaS product that interfaces with a university's legacy Student Information System (SIS) through a well-defined sidecar integration architecture. It does not modify or replace existing academic databases. Instead, it extends institutional capability by importing foundational identity data (student registration numbers, names, course enrolment records) from the SIS using a daily automated import pipeline, and exports verified attendance percentage summaries back to the SIS for academic record-keeping, examination eligibility determinations, and payroll confirmation for teaching staff.

The system is architected around a decentralised edge computing model. During live academic sessions, no active internet connection is required in the classroom. All validation, session management, and attendance logging operations execute locally on the Coordinator's PWA device. Data is transmitted to the central cloud infrastructure only during scheduled synchronisation windows, using the Chunked Resume-Sync protocol to ensure complete data integrity regardless of network quality.

As a multi-tenant platform, QAAT supports simultaneous operation across multiple independent universities, each configured with unique institutional branding, academic hierarchies, attendance threshold policies, proximity sensitivity calibrations, and data retention rules. Tenant data is logically isolated at the database level using Row-Level Security (RLS) policies, ensuring that no institution can access data belonging to another.

2.2 System Context Diagram (Narrative)

The QAAT system exists at the intersection of three operational environments. The first is the Physical Classroom Environment, characterised by unreliable or absent internet connectivity, containing the Coordinator's PWA device, BLE Beacons, the Lecturer, and Students with their university-issued QR credentials. The second is the Institutional Administrative Environment, comprising the offices of the Vice Chancellor, Director of Quality Assurance, Quality Assurance Officers, and Coordinators, all connected to stable campus Wi-Fi or ethernet. The third is the QAAT Cloud Infrastructure, which hosts the multi-tenant PostgreSQL database, the Golang API gateway, QR generation microservices, and automated reporting engines.

The Coordinator's PWA functions as the critical bridge between the Physical Classroom Environment and the Cloud Infrastructure, collecting offline data and transmitting it when connectivity is available.

2.3 Product Functions — High-Level Summary

The QAAT system provides the following primary categories of functionality:

#	Function Category	Summary
F-01	**Identity Issuance**	Generation and email delivery of RSA-signed, institution-specific QR credentials to enrolled students and Lecturers.

F-02	**Session Lifecycle Management**	Creation, activation, monitoring, and closure of lecture sessions including automated temporal locking.
F-03	**Attendance Verification**	Multi-factor attendance confirmation combining QR cryptographic validation, hardware device fingerprinting, and BLE beacon proximity enforcement.
F-04	**Offline Edge Computing**	Local session hosting, roster management, and attendance logging via the Coordinator PWA without internet dependency.
F-05	**Secure Data Synchronisation**	Chunked, resumable, AES-256 encrypted transmission of session logs from the PWA Edge Server to the central cloud database.
F-06	**Absence and Coordinator Delegation**	Emergency session management protocols including Warden delegation and automatic session recovery when the primary Coordinator is unavailable.
F-07	**Lecturer Attendance Tracking**	Separate, authoritative recording of Lecturer session start and end times, contact hours, and punctuality metrics.
F-08	**Multi-Role Dashboard**	Role-differentiated analytical dashboards for the Vice Chancellor, DQA Director, QA Officers, and Coordinators.
F-09	**Examination Eligibility**	Automated computation and enforcement of the institutional attendance threshold, generating real-time eligibility statuses and cryptographically signed exam clearance tokens.
F-10	**Legacy System Integration**	Sidecar schema integration with university SIS databases via configurable import/export pipelines and RESTful API bridges.
F-11	**Multi-Tenant Policy Engine**	Per-tenant configuration of attendance thresholds, proximity sensitivity, auto-close durations, academic calendar mappings, and institutional branding.
F-12	**Fraud Detection and Anomaly Reporting**	Automated detection of proxy scanning, device collisions, temporal manipulation, and ghost sessions with DQA Dashboard alerting.

2.4 User Classes and Characteristics

2.4.1 Vice Chancellor (VC)

The VC is the highest institutional authority. Technical proficiency: low-to-moderate. Interaction mode: web browser on desktop or tablet, always online. Primary need: high-level institutional performance intelligence, audit trails, and examination eligibility enforcement. Access level: read-only, full granularity with appetite-based filtering across all institutional data.

2.4.2 Director of Quality Assurance (DQA)

The DQA Director is the policy authority for QAAT configurations within their institution. Technical proficiency: moderate. Interaction mode: web dashboard. Primary need: aggregate attendance trends, threshold policy management, and ghost lecture detection. Access level: read-write for institutional policy settings; read-only for raw attendance logs.

2.4.3 Quality Assurance Officer (QA Officer)

The QA Officer performs field-level auditing and exception handling. Technical proficiency: moderate. Interaction mode: web dashboard, mobile browser. Primary need: real-time session monitoring, hardware binding resets, manual audit interventions, and spot-check verification. Access level: read-write for exception management within their assigned scope.

2.4.4 Course Coordinator

The Coordinator is the sole holder of the Coordinator PWA application. Technical proficiency: moderate. Interaction mode: PWA installed on a dedicated or personal smartphone. Primary need: session initialisation, Lecturer gating, local data custody, and synchronisation. Access level: operational control over sessions for their assigned Course; zero read access to attendance percentages or individual student records.

2.4.5 Lecturer

The Lecturer participates in sessions but does not install any dedicated application. Technical proficiency: low. Interaction mode: native smartphone camera (Google Lens, iOS Camera). Primary need: scanning the Coordinator PWA QR to gate-open a session and scanning to close it. Access level: no dashboard access; participates only through the physical scan interaction.

2.4.6 Student

Students are passive credential holders. Technical proficiency: any level. Interaction mode: native smartphone camera to scan their university-issued personal QR code. Primary need: to mark attendance by scanning their QR code in proximity to the Coordinator PWA or Warden device. Access level: personal attendance percentage view only, via a lightweight status portal.

2.4.7 Session Warden (Delegated Student)

The Warden is a trusted student temporarily assigned by the Coordinator to act as the local session host during the Coordinator's absence. Technical proficiency: moderate. Interaction mode: browser-based link portal. Primary need: to host the local session, act as the BLE proximity anchor, and trigger data submission. Access level: operational control limited to the specific delegated session; no access to any other data.

2.4.8 System Administrator (MR. LIGHT / Platform Operator)

The Platform Administrator manages the underlying cloud infrastructure, tenant onboarding, system health monitoring, and global security updates. Technical proficiency: expert. Access level: superadmin across all tenants.

2.5 Operating Environment

2.5.1 Coordinator PWA

- Runtime: Any modern browser (Chrome 90+, Safari 14+, Firefox 90+) on Android 9+, iOS 14+, or Windows 10+.
- Hardware minimum: 3 GB RAM, Bluetooth 4.0+ for BLE beacon detection, Wi-Fi 802.11n/ac for LAN hosting.
- Storage: Minimum 500 MB available browser storage for IndexedDB session vault.
- Network: Operates fully offline during sessions; requires internet for initial daily fetch and post-session atomic sync.

2.5.2 Student and Lecturer Devices

- Any smartphone or tablet with a functioning camera and the ability to open a URL in a browser.
- No application installation required. No minimum OS version beyond basic browser URL support.

2.5.3 Administrative Dashboards

- Web browser (Chrome 90+, Edge 90+, Firefox 90+, Safari 14+) on desktop or mobile.
- Stable internet connection required (minimum 1 Mbps).

2.5.4 Backend Infrastructure

- Runtime: Go (Golang) 1.21+ for API gateway; Node.js 20+ for QR generation microservice.
- Database: PostgreSQL 15+ with Row-Level Security enabled.
- Deployment: Containerised via Docker/Kubernetes on a cloud platform (AWS, GCP, or Azure) with multi-region failover support.
- Cache Layer: Redis 7+ for session tokens and Daily Manifest caching.

2.6 Design and Implementation Constraints

10. The system MUST operate as an offline-first architecture. All attendance validation logic MUST execute locally on the Coordinator's PWA without any external API calls during the session.
11. No student or Lecturer is required to install any dedicated mobile application. All student-facing and Lecturer-facing interactions MUST be achievable through native device cameras and standard web browsers.
12. The Coordinator's PWA MUST be distributable without listing on the Apple App Store or Google Play Store, implemented as an installable Progressive Web App via browser 'Add to Home Screen' functionality.
13. The system MUST support the existing university student database without requiring re-registration of students. Identity import MUST be achievable via CSV upload or a standard REST API integration.
14. All data transmission between the Coordinator PWA and the central server MUST be encrypted using TLS 1.3 and AES-256.
15. Student personal data (names, registration numbers) MUST NOT be stored in plain text on the Coordinator's local device. Only cryptographic hashes and session tokens may be persisted locally.
16. The system MUST be culturally and technically adaptable for deployment across multiple universities with differing academic calendar structures, attendance policies, and branding identities.

2.7 Assumptions and Dependencies

2.7.1 Assumptions

- Each university deploying QAAT designates one Coordinator per Course who is responsible for all sessions under that Course.
- Students enrolled in a Course are, by default, enrolled in all Course Units within that Course. Exceptions are managed through a per-student exclusion list maintained by the DQA.
- BLE Beacons will be procured, installed, and maintained by the university's facilities team in designated lecture halls prior to QAAT deployment.
- Each student possesses a university-issued email account and a smartphone capable of scanning QR codes.
- The university's existing SIS can export student records in CSV format or expose them via a REST API for the QAAT import pipeline.
- Coordinators are university staff members with a legitimate institutional email address and are issued dedicated smartphones or tablets for QAAT operations.

2.7.2 Dependencies

- Availability of the university's SIS for initial data import and periodic synchronisation.
- Availability of an institutional SMTP mail relay or third-party email service (e.g., SendGrid) for QR code distribution.
- Continuous power supply to BLE beacons in lecture halls. Battery replacement schedule must be maintained by facilities.
- Browser API support for Web Bluetooth (navigator.bluetooth) on Coordinator PWA devices.

3. Specific Requirements

3.1 Functional Requirements

3.1.1 FR-01: Identity Issuance and QR Code Management

FR-01.1 Student QR Code Generation

The system SHALL generate a unique, RSA-2048-signed QR code for every student imported into the QAAT platform. Each QR code SHALL encode a signed JSON payload containing: the Student Registration Number, the Institution Tenant ID, the Academic Year, a Cryptographic Serial Number, an Expiry Date (end of academic year), and an HMAC integrity checksum. The private key used for signing SHALL be unique to the institution's tenant configuration and rotated annually.

FR-01.2 QR Code Delivery

The system SHALL deliver each student's QR code to their registered university email address as a high-resolution PNG image attachment (minimum 1024x1024 pixels) within 24 hours of account creation or the start of a new academic year. The email SHALL originate from an institutionally branded address and SHALL include instructions for saving and using the QR code.

FR-01.3 QR Code Uniqueness and Validity

Each QR code SHALL be uniquely bound to one Student Registration Number. The system SHALL reject any QR code presented during a session that does not pass the RSA signature verification against the institution's Public Key. The system SHALL reject QR codes with an Expiry Date earlier than the current session date. The system SHALL reject QR codes belonging to a different institution's Tenant ID than the one configured on the Coordinator's PWA for the active session.

FR-01.4 QR Code Reissuance

The system SHALL provide a QR code reissuance function accessible to QA Officers. Reissuance SHALL invalidate the previous QR code by rotating its Cryptographic Serial Number. The system SHALL log all reissuance events with a timestamp, the requesting QA Officer's identity, and the reason code.

3.1.2 FR-02: Coordinator PWA — Session Management

FR-02.1 Daily Manifest Fetch

The Coordinator's PWA SHALL support a 'Daily Fetch' operation, executable when an internet connection is available. This operation SHALL download from the central server: (a) the complete, hashed student roster for all Course Units assigned to the Coordinator for that academic day; (b) the BLE Beacon Room IDs for the Coordinator's assigned venues; (c) the current institutional policy parameters including attendance threshold, session duration limits, and proximity RSSI threshold; and (d) the current institution Public Key for QR signature verification. All downloaded data SHALL be encrypted and stored in the PWA's IndexedDB.

FR-02.2 Session Initialisation

The PWA SHALL present the Coordinator with a dropdown-only session configuration interface. The Coordinator SHALL select the Course Unit from a list pre-filtered to their assigned units. The PWA SHALL not permit free-text entry of Course Unit, Venue, or any session parameter. A session SHALL be initialised only when: (a) a valid Course Unit is selected, (b) the BLE Beacon for the assigned room is detected with a signal strength meeting the institutional RSSI threshold, and (c) no other session is currently in ACTIVE state on the device.

FR-02.3 Single Active Session Constraint

The PWA SHALL implement a strict State Machine that prevents more than one session from being in ACTIVE state simultaneously. If the Coordinator attempts to initialise a new session while another is active, the PWA SHALL display an error message and SHALL NOT create a new session. The Coordinator MUST end the active session before a new one can begin.

FR-02.4 Session Closure

The PWA SHALL provide a 'End Session' button available only to the authenticated Coordinator. Pressing this button SHALL transition the session to CLOSED state, cease acceptance of new attendance records, record the session end time, and place the session log in the Synchronisation Queue. The PWA SHALL seal the session log with an AES-256 encrypted, JWT-authenticated package before queuing.

FR-02.5 Automated Session Expiry

The PWA SHALL implement an automated session expiry timer. The session SHALL automatically transition to CLOSED state under the following conditions: (a) the student check-in window expires at T+120 minutes from the Lecturer gate-open event; (b) the total session duration reaches T+180 minutes from initialisation, regardless of Coordinator action. Auto-closed sessions SHALL be flagged with a status code of AUTO_CLOSED in the session log for DQA Dashboard review.

FR-02.6 Multi-Session Daily Operation

The PWA SHALL support multiple sequential sessions within a single operating day. Each closed session SHALL be individually packaged, sealed, and appended to the Synchronisation Queue. The Coordinator SHALL be able to initiate a new session immediately upon closing the previous one without requiring a device restart.

3.1.3 FR-03: Lecturer Gating — Sequential Session Activation

FR-03.1 Mandatory Lecturer Gate-Open

The PWA SHALL generate a unique one-time Gate-Open QR code upon session initialisation. This Gate-Open QR code SHALL expire after 15 minutes. The PWA SHALL NOT accept any student attendance scan until the Lecturer has presented this Gate-Open QR code using their native device camera, and the PWA has verified the Lecturer's Staff QR code in response. The session SHALL remain in PENDING_LECTURER state, displaying a 'Waiting for Lecturer' status screen, until this handshake is complete.

FR-03.2 Lecturer Identification

The Lecturer's Staff QR code, issued by the university administration and delivered to their institutional email, SHALL encode the Lecturer's Staff ID, their assigned Course Units for the current semester, their Tenant ID, and an RSA signature. The PWA SHALL verify the Lecturer's QR against the locally cached roster to confirm the Lecturer is authorised for the specific Course Unit being taught.

FR-03.3 Lecturer Gate-Close

At the end of the session, the Coordinator SHALL request the Lecturer to scan the Coordinator's 'Gate-Close' QR code to officially record the end of the teaching period. If the Lecturer does not perform the Gate-Close scan, the session end time SHALL be recorded as the automated expiry time. Sessions with a missing Lecturer Gate-Close event SHALL be flagged as INCOMPLETE_LECTURER_LOG for QA Officer review.

FR-03.4 Lecturer Attendance Record

The system SHALL maintain a separate Lecturer Attendance Log, independent of the Student Attendance Log, capturing: Lecturer Staff ID, Session ID, Gate-Open timestamp, Gate-Close timestamp, Calculated Contact Hours (Gate-Close minus Gate-Open), assigned Course Unit, venue, and session date. This log SHALL be accessible only to QA Officers, DQA Directors, and the Vice Chancellor. Coordinators SHALL NOT have read access to Lecturer Attendance Logs.

3.1.4 FR-04: Student Attendance Verification

FR-04.1 Student Self-Scan Protocol

Students SHALL initiate attendance by scanning their university-issued personal QR code using their native device camera application. The scan SHALL open a session-specific URL hosted locally on the Coordinator's PWA LAN. The PWA SHALL perform the following validation sequence in order: (1) RSA signature verification of the student's QR code; (2) Student Registration Number lookup against the locally cached Course Unit roster; (3) BLE Beacon proximity check; (4) Hardware device fingerprint verification; and (5) Duplicate scan check for the current session.

FR-04.2 BLE Beacon Proximity Enforcement

The system SHALL only accept an attendance record if the student's device detects the room's assigned BLE Beacon with a signal strength (RSSI) equal to or greater than the institution-configured proximity threshold (default: -65 dBm). The system SHALL use a weighted 10-second rolling average of RSSI readings to prevent transient signal fluctuations from causing false rejections. The system SHALL reject attendance requests where the Beacon is not detected or the RSSI is below threshold, displaying a 'Move closer to the session anchor' message.

FR-04.3 Hardware Device Fingerprint Binding

On a student's first ever attendance scan within the platform, the system SHALL capture a synthetic hardware fingerprint computed from: browser User-Agent string, screen resolution, colour depth, device pixel ratio, CPU logical core count, and a canvas rendering hash. This fingerprint SHALL be bound to the student's Registration Number in the Hardware Vault table. On all subsequent scans, the system SHALL recompute the fingerprint and compare it against the stored value. A mismatch SHALL result in rejection and a DEVICE_MISMATCH flag logged for QA Officer review.

FR-04.4 One-Scan-Per-Session Enforcement

The system SHALL enforce a strict one-scan-per-session rule. Once a student's Registration Number is logged as PRESENT for a given Session ID, any subsequent scan of the same Registration Number within the same session SHALL be silently rejected. The system SHALL also enforce a one-device-per-session rule: once a hardware fingerprint has been used to register one student's attendance in a session, the same fingerprint SHALL NOT be accepted for any other student's Registration Number within that same session.

FR-04.5 Real-Time Student Feedback

Upon successful attendance verification, the system SHALL display an immediate confirmation screen on the student's device showing: the Course Unit name, Lecturer name, venue, session date and time, and the student's current personal attendance percentage for that Course Unit computed against the institutional threshold. If the student's percentage is below the institutional threshold, the confirmation screen SHALL display a prominent amber or red warning with the specific deficit (e.g., 'You are 5 sessions below the required threshold').

3.1.5 FR-05: Offline Data Persistence and Synchronisation

FR-05.1 Local Data Vault

The PWA SHALL store all session logs, attendance records, and hardware fingerprint bindings in the browser's IndexedDB. The IndexedDB store SHALL use AES-256 encryption at rest, keyed with a device-specific secret. All stored records SHALL include a unique Record ID, a creation timestamp, a synchronisation status flag (PENDING, SYNCED, FAILED), and a cryptographic integrity checksum.

FR-05.2 Synchronisation Queue Management

Upon session closure, the sealed session package SHALL be appended to an Outbox Queue persisted in IndexedDB. The PWA Service Worker SHALL continuously monitor for internet connectivity. Upon detecting a stable connection, the Service Worker SHALL begin processing the Outbox Queue, transmitting one session package at a time.

FR-05.3 Chunked Resume-on-Reconnect Protocol

Each session package in the Outbox Queue SHALL be uploaded as a series of encrypted JSON chunks. For each chunk, the central server SHALL return a cryptographically signed acknowledgement receipt. The PWA SHALL only remove a chunk from the queue upon receipt of this acknowledgement. If the connection is interrupted during upload, the PWA SHALL record the index of the last acknowledged chunk. Upon reconnection, the PWA SHALL resume transmission from the next unacknowledged chunk, ensuring zero data loss and preventing duplicate records.

FR-05.4 7-Day Local Archive

The PWA SHALL maintain a 7-day rolling archive of all session packages in IndexedDB, regardless of synchronisation status. Packages older than 7 days that have been successfully synchronised SHALL be purged from local storage. Packages older than 7 days that have NOT been synchronised SHALL be escalated with a SYNC_OVERDUE alert to the Coordinator and QA Officer dashboards.

FR-05.5 Server-Side Deduplication

Upon receiving a synchronisation package, the central server SHALL perform deduplication using Vector Clock comparison (Session ID + Record ID + Coordinator ID) before writing to the production database. Records that fail deduplication SHALL be logged as DUPLICATE_REJECTED and reported in the QA Officer dashboard without blocking the remainder of the batch.

3.1.6 FR-06: Absence and Warden Delegation

FR-06.1 Coordinator Absence Protocol

When the primary Coordinator is unable to attend a scheduled session, they SHALL be able to generate a one-time, session-specific Warden Delegation Link via the PWA while still online. This link SHALL be cryptographically bound to: the designated Warden's Student Registration Number, the Session ID, the Course Unit ID, the venue ID, and a session time window.

FR-06.2 Warden Link Activation

Upon clicking the Warden Delegation Link in their browser, the Warden's device SHALL be elevated to function as a Temporary Edge Server for the designated session only. The system SHALL perform a GPS geofence check to verify the Warden's device is located within the registered coordinates of the designated lecture venue. If geofence verification fails, the link SHALL NOT activate. The Warden's device SHALL NOT have access to any session data beyond the current delegated session.

FR-06.3 Warden Session Execution

Once activated, the Warden Delegation Link portal SHALL: display a Gate-Open QR code for Lecturer scanning; act as the BLE proximity anchor for student attendance (if the Warden's device supports Web Bluetooth); capture all student attendance in the browser's local cache; and trigger an auto-submission of the encrypted session package 5 minutes after the session's scheduled end time, or immediately upon the Warden manually pressing 'Submit Session'.

FR-06.4 Data Custody Transfer

Data collected by the Warden SHALL be delivered to the Coordinator's PWA account via the central server sync pipeline upon submission. The Coordinator SHALL receive a push notification ('Session [ID] data received from Warden [Name]') and SHALL be required to review and approve the session data before it is committed as final. The VC Dashboard SHALL flag such sessions with a WARDEN_VERIFIED badge.

FR-06.5 Extended Absence — Same-Day Multiple Sessions

If the Coordinator is absent for an entire day during which multiple sessions are scheduled, the DQA Officer SHALL have the authority to generate multiple Warden Delegation Links for different sessions using the administrative dashboard. Each Delegation Link SHALL be independently cryptographically isolated.

3.1.7 FR-07: Multi-Role Administrative Dashboards

FR-07.1 Vice Chancellor Dashboard

The Vice Chancellor's dashboard SHALL provide full granular read access to all institutional attendance data with the following capabilities:

- Appetite-based filtering: by School, College, Department, Course, Course Unit, Lecturer, Student, Academic Year, Semester, Intake, Session Type (Day / Evening / Weekend / E-Learning), and date range.
- Institutional Efficiency Ratio (IER) display: calculated as (Actual Contact Hours / Scheduled Contact Hours) x Average Student Attendance Percentage.
- Ghost lecture flagging: automatic identification of sessions where Lecturer Gate-Open was recorded but student attendance was below 5% of the enrolled class.

- Lecturer workload profile: across all Course Units taught, with comparison between scheduled and actual contact hours.
- One-click print/export of filtered data as a formatted PDF Audit Report.

FR-07.2 DQA Director Dashboard

The DQA Director's dashboard SHALL provide summary-level institutional analytics and policy management with the following capabilities:

- Global threshold adjustment: ability to change the institutional attendance percentage threshold (0%–100%), with immediate system-wide propagation.
- Aggregated course health: heatmap view of attendance compliance per course unit and department.
- Automated exam eligibility list: all students below threshold flagged with specific deficit details.
- Attendance trend analysis: week-over-week and month-over-month graphs per department.
- Lecturer punctuality summary: aggregate view of late start frequency per Lecturer.

FR-07.3 Quality Assurance Officer Dashboard

The QA Officer's dashboard SHALL provide real-time field auditing capabilities with the following functions:

- Live session monitor: display of currently active sessions, their Coordinators, and real-time student count.
- Anomaly alert feed: real-time alerts for DEVICE_MISMATCH, DUPLICATE_SCAN_ATTEMPT, SYNC_OVERDUE, and AUTO_CLOSED flags.
- Hardware binding management: ability to reset a student's device fingerprint binding (limited to 2 resets per student per semester, with mandatory reason code entry).
- Manual attendance correction: ability to add an ADJUSTMENT entry to a session log with full audit trail (original log preserved; adjustment logged separately).
- Coordinator health metrics: sync latency and manual-override ratio monitoring.

FR-07.4 Coordinator PWA Operational View

The Coordinator's PWA SHALL display only the following information: current session status (IDLE, PENDING_LECTURER, ACTIVE, CLOSED), live count of students checked in for the current session, a searchable student list limited to names and check-in status (PRESENT / NOT YET SCANNED), synchronisation queue status (count of pending sessions and last successful sync timestamp). The Coordinator SHALL NOT be able to access attendance percentages, historical session data, or Lecturer attendance records.

3.1.8 FR-08: Examination Eligibility Management

FR-08.1 Real-Time Eligibility Computation

The system SHALL continuously compute each student's attendance percentage per Course Unit, defined as (Sessions Attended / Total Sessions Held to Date) x 100. This computation SHALL occur server-side after each Atomic Sync and SHALL be reflected in the student's status portal within 60 seconds of a successful sync.

FR-08.2 Threshold Enforcement

The system SHALL compare each student's computed attendance percentage against the institution's current DQA-configured threshold. Students falling below the threshold SHALL be assigned status EXAM_INELIGIBLE for the affected Course Unit. The system SHALL generate an automated eligibility report accessible to DQA Officers and the Registrar, exportable in CSV and PDF formats.

FR-08.3 Cryptographically Signed Exam Clearance Token

For examination administration, the system SHALL generate a cryptographically signed, server-issued Exam Clearance QR code for each eligible student per Course Unit. This token SHALL encode: Student Registration Number, Course Unit, current attendance percentage, threshold requirement, eligibility status, and an RSA signature. Exam invigilators SHALL scan this token to verify eligibility without requiring access to the full QAAT dashboard.

FR-08.4 Student Status Portal

The system SHALL provide a lightweight, low-bandwidth student status portal accessible via a simple web URL or from the student's saved PWA shortcut. This portal SHALL display the student's attendance percentage, status (ELIGIBLE or INELIGIBLE), and the applicable threshold per Course Unit. Data SHALL be fetched from the server when online; a cached copy SHALL be displayed offline with a 'Last updated' timestamp.

3.1.9 FR-09: Legacy System Integration

FR-09.1 Import Pipeline

The system SHALL support two modes of student data importation from the university's existing SIS: (a) Manual CSV Upload — the DQA Officer uploads a standardised CSV file containing Student Registration Number, Full Name, Email Address, Course ID, Academic Year, Semester, Intake Session, and Enrollment Status; (b) Automated API Pull — a configurable nightly CRON job that queries the SIS REST API for new and updated student records. All imported student records SHALL be inserted or updated in the QAAT Students table using the Student Registration Number as the unique key.

FR-09.2 Export Pipeline

Upon request by the DQA Officer or automatically at the end of each semester, the system SHALL export final attendance percentages per student per Course Unit back to the university SIS in a format specified by the institution (CSV, JSON, or REST API push). The export SHALL include only the Student Registration Number, Course Unit Code, Attendance Percentage, Eligibility Status, and the reporting period.

FR-09.3 Sidecar Schema Architecture

The QAAT database SHALL implement a Sidecar Schema that is logically separate from the university SIS. The Student Registration Number SHALL serve as the foreign key linking QAAT records to the SIS master record. The QAAT schema SHALL not modify, overwrite, or delete any records in the university SIS. All QAAT-specific extended fields (hardware fingerprints, session logs, beacon data) SHALL reside exclusively in the QAAT Sidecar Schema.

3.1.10 FR-10: Multi-Tenant Policy Engine

FR-10.1 Tenant Isolation

Each university tenant SHALL be assigned a unique Tenant ID. All database tables SHALL include a tenant_id column, and PostgreSQL Row-Level Security (RLS) policies SHALL be applied to enforce data isolation. No query from one tenant's authenticated session SHALL return records belonging to another tenant.

FR-10.2 Per-Tenant Policy Configuration

Each tenant's DQA Director SHALL be able to configure the following parameters through the DQA Dashboard: attendance percentage threshold (default 75%), student check-in window duration (default 120 minutes, range 30–180 minutes), session auto-kill duration (default 180 minutes, range 60–360 minutes), BLE proximity RSSI threshold (default -65 dBm), institutional branding (logo, name, colour palette for QR codes and student portal), registration number prefix format, and academic calendar (semesters, intakes, year structure).

FR-10.3 Policy Propagation

Changes to tenant policy parameters SHALL be propagated to all Coordinator PWAs associated with the tenant during their next Daily Fetch operation. Policy changes SHALL not take effect mid-session; they SHALL apply to all new sessions initiated after the next Daily Fetch.

3.2 Non-Functional Requirements

3.2.1 NFR-01: Performance

ID	Metric	Target
NFR-01.1	Student QR scan validation latency	< 800 milliseconds end-to-end on the Coordinator PWA (local processing, no network round-trip).
NFR-01.2	Session initialisation time	< 3 seconds from Course Unit selection to session ACTIVE state.
NFR-01.3	Daily Fetch duration	< 5 seconds for roster of up to 2,000 students on a 10 Mbps connection.
NFR-01.4	Atomic Sync throughput	Full session log of 500 attendance records uploaded in < 30 seconds on a 5 Mbps connection.
NFR-01.5	Dashboard data load time	< 4 seconds for filtered VC/DQA reports covering up to 10,000 records.
NFR-01.6	Peak concurrent scan handling	The PWA MUST handle up to 300 simultaneous student scans per minute without data loss or session corruption.

3.2.2 NFR-02: Reliability and Availability

- The central cloud infrastructure SHALL achieve 99.9% monthly uptime (maximum 43.8 minutes unplanned downtime per month).
- The Coordinator PWA SHALL operate at 100% functional availability in offline mode. Internet connectivity SHALL not be a prerequisite for any attendance verification operation.
- No session data SHALL be lost due to device failure if the device has been operational for at least 30 seconds after the scan. The 7-day local archive SHALL protect against catastrophic device loss.
- The Chunked Resume-Sync protocol SHALL achieve a minimum of 99.99% data delivery reliability across all synchronisation operations.

3.2.3 NFR-03: Security

- All QR codes SHALL be signed using RSA-2048 with SHA-256 hashing. Private keys SHALL be stored in a Hardware Security Module (HSM) or equivalent software vault, never in application code.
- All data transmitted between the Coordinator PWA and the central server SHALL use TLS 1.3. TLS 1.2 SHALL be supported for legacy compatibility but with a deprecation notice.
- All data at rest in the central PostgreSQL database SHALL be encrypted at the filesystem level using AES-256.
- All data at rest in the Coordinator's IndexedDB SHALL be encrypted using AES-256, keyed with a device-specific, server-issued secret.

- JWT tokens issued to the Coordinator PWA SHALL have a maximum lifetime of 24 hours and SHALL include a jti (JWT ID) claim for replay prevention.
- The system SHALL implement rate limiting on all API endpoints. The attendance submission endpoint SHALL be limited to 50 requests per second per Coordinator device.
- All administrative dashboard authentication SHALL support multi-factor authentication (MFA). TOTP-based MFA SHALL be mandatory for VC and DQA Director accounts.
- Device fingerprint data SHALL not be used to identify a student independently of their Registration Number. It SHALL only serve as a session-scoped anti-proxy mechanism.

3.2.4 NFR-04: Usability

- The student check-in process SHALL be completable in under 10 seconds from QR code scan to confirmation screen display.
- The Coordinator PWA session setup SHALL require no free-text entry. All inputs SHALL be accomplished via dropdown selections populated from the Daily Manifest.
- All error messages SHALL be written in plain language, identifying the cause and recommended action. Technical error codes SHALL be displayed only in a collapsible 'Details' section.
- The student status portal SHALL load in under 3 seconds on a 2G mobile connection (equivalent to approximately 250 kbps throughput).
- The PWA SHALL be fully navigable without any prior training for the student (scan QR, view confirmation). Coordinator-level training SHALL be completable in under 2 hours.
- All dashboard interfaces SHALL comply with WCAG 2.1 Level AA accessibility standards.

3.2.5 NFR-05: Maintainability and Extensibility

- The system SHALL be architectured as a set of loosely coupled microservices to allow independent scaling and deployment of the QR generation, authentication, attendance processing, and reporting components.
- All tenant policy parameters SHALL be stored as configurable database records, not hardcoded values, enabling zero-downtime policy updates.
- The PWA SHALL be updated automatically via Service Worker cache replacement. Coordinators SHALL not need to manually reinstall the PWA to receive updates.
- The central API SHALL be versioned (e.g., /api/v1/, /api/v2/) to maintain backward compatibility during system upgrades.
- The system SHALL provide a comprehensive internal logging and tracing infrastructure (e.g., structured JSON logs with correlation IDs) to facilitate debugging without requiring direct database access.

3.2.6 NFR-06: Scalability

- The central database SHALL be designed for horizontal sharding by tenant_id to accommodate growth to over 500 tenant universities and 1 million student records without degradation of query performance.

- The Daily Manifest delivery infrastructure SHALL utilise a CDN with edge caching to ensure < 5-second fetch times regardless of Coordinator geographic location.
- The system architecture SHALL support auto-scaling of API gateway instances to accommodate peak loads of 10,000 concurrent sync operations.
- The PWA local processing architecture SHALL handle lecture halls of up to 500 concurrent students with acceptable scan latency.

4. External Interface Requirements

4.1 User Interfaces

4.1.1 Coordinator PWA Interface

The Coordinator PWA SHALL present a minimal, task-oriented interface with the following primary screens: (1) Daily Fetch Screen — displays sync status, last fetch timestamp, and a 'Sync Now' button; (2) Session Setup Screen — dropdown menus for Course Unit and Venue selection; (3) Session Active Screen — displays session status, Lecturer verification state, live attendance count, and a searchable student presence list; (4) End Session Screen — confirmation dialog with session summary before closure; (5) Sync Queue Screen — list of sessions pending upload with status indicators.

The interface SHALL use the institution's configured colour palette. Primary actions (Start Session, End Session, Sync) SHALL use high-contrast, touch-friendly buttons (minimum 44x44 px touch target per WCAG 2.1 guidelines).

4.1.2 Student Confirmation Interface

The student-facing confirmation interface SHALL be a lightweight, browser-rendered single-page displaying: institution logo, a large status icon (green checkmark for success, red cross for failure), a brief status message, the session details (Course Unit, Lecturer, Venue, Date, Time), and the student's real-time attendance percentage with threshold comparison. The interface SHALL render correctly on screens as small as 320px width.

4.1.3 Administrative Dashboards

All administrative dashboards SHALL be responsive web applications compatible with modern desktop and mobile browsers. They SHALL provide data table views with sortable and filterable columns, chart visualisations (bar, line, and heatmap), and PDF export functionality. All charts SHALL include accessible text alternatives.

4.2 Hardware Interfaces

4.2.1 BLE Beacon Interface

The system SHALL be compatible with BLE Beacons supporting the iBeacon or Eddystone-UID advertising formats. Each Beacon SHALL be assigned a unique UUID registered in the QAAT Venue Registry. The Coordinator PWA SHALL use the Web Bluetooth API (navigator.bluetooth.requestLEScan) to detect and read Beacon advertising data. Beacon firmware SHALL support battery level reporting via the BLE Battery Service (0x180F) characteristic for predictive maintenance monitoring.

4.3 Software Interfaces

4.3.1 University SIS Integration API

The QAAT import pipeline SHALL interface with the university SIS via a RESTful API supporting OAuth 2.0 authentication. The expected SIS API endpoints are: GET /students?course_id={id}&academic_year={year} — returns enrolled student records; GET /courses — returns the institutional course hierarchy. If the SIS does not expose an API, the system SHALL accept a standardised CSV upload. The CSV format SHALL be documented in Appendix A.

4.3.2 Email Service Interface

The QR code distribution service SHALL interface with SMTP relay servers or REST-based email APIs (compatible with SendGrid, Mailgun, or AWS SES API formats) for bulk email delivery. The system SHALL support delivery rate limiting (maximum 100 emails/second) and bounce handling.

4.3.3 PWA Service Worker Interface

The Coordinator PWA SHALL utilise the browser Service Worker API (W3C Service Workers Level 1) for background synchronisation via the Background Sync API and for offline cache management via the Cache API.

4.4 Communication Interfaces

All communications between the Coordinator PWA and the central API SHALL use HTTPS (TLS 1.3) on port 443. The local LAN session between student devices and the Coordinator PWA SHALL use HTTP on port 8080 within the local subnet only; no outbound routing of this traffic occurs. BLE communications between beacons and the Coordinator PWA SHALL be Bluetooth 4.2 or higher, operating in the 2.4 GHz ISM band.

5. System Quality Attributes

5.1 Security and Privacy

The QAAT system is designed with a Privacy-by-Design philosophy. Student names and personally identifiable information (PII) are not transmitted or stored on the Coordinator's local device. Only Student Registration Numbers (opaque identifiers in the classroom context) and their SHA-256 hashed device fingerprints are persisted locally. Full PII is reconstituted only within the encrypted central database behind authenticated API access. The system is designed to comply with applicable data protection regulations including GDPR (where applicable), and each institution's own data governance policies.

5.2 Fault Tolerance

The system implements multiple fault tolerance mechanisms. The edge computing architecture ensures that a central server outage has zero impact on live session operation. The 7-day local archive provides a recovery window for scenarios involving Coordinator device loss or failure. The Chunked Resume-Sync protocol prevents data loss from intermittent network failures. The 3-hour auto-kill prevents indefinitely open sessions from corrupting the audit trail.

5.3 Auditability

Every system action is logged with an immutable timestamp, the identity of the actor, and the nature of the action. Attendance logs are append-only: records may not be deleted from production tables. Adjustments to historical records create supplementary ADJUSTMENT entries that are visible in audit trails. All administrative actions on the QA Officer and DQA Director dashboards are logged in the Admin Audit Log, accessible exclusively to MR. LIGHT's Platform Administrator account and the VC for compliance review.

5.4 Interoperability

The sidecar architecture ensures QAAT integrates with university systems without disruption. The use of standard data formats (JSON, CSV), open authentication standards (OAuth 2.0, JWT), and versioned REST APIs ensures compatibility with a wide range of university SIS platforms and administrative tooling. The multi-tenant design allows QAAT to serve institutions running different SIS platforms simultaneously.

6. Use Case Descriptions

6.1 UC-01: Conduct Standard Lecture Session (Coordinator Present)

Field	Description
Use Case ID	UC-01
Name	Conduct Standard Lecture Session
Actors	Coordinator (primary), Lecturer, Students
Preconditions	1. Daily Manifest has been fetched. 2. BLE Beacon is operational in the assigned room. 3. Lecturer's Staff QR code is available.
Main Flow	1. Coordinator opens PWA, selects Course Unit. 2. System verifies BLE Beacon presence. 3. Session initialised (PENDING_LECTURER state). 4. Lecturer scans Gate-Open QR. 5. Session transitions to ACTIVE. 6. Students scan personal QR codes; proximity and fingerprint verified. 7. Session auto-closes after student window (T+120 min). 8. Lecturer scans Gate-Close QR. 9. Coordinator ends session manually. 10. Session sealed and placed in Sync Queue.
Alternative Flows	A1: Student QR fails signature check → rejected with message. A2: Student device fingerprint mismatch → rejected, DEVICE_MISMATCH flag logged. A3: Coordinator forgets to end session → auto-kill at T+180 min, AUTO_CLOSED flag set.
Postconditions	Session log sealed, placed in Sync Queue, Lecturer attendance record created, student attendance percentages queued for update on next sync.

6.2 UC-02: Coordinator Absent — Warden Delegation

Field	Description
Use Case ID	UC-02
Name	Coordinator Absent — Warden Delegation
Actors	Coordinator (remote), Warden (Student), Lecturer, Students
Preconditions	Coordinator has internet access to generate Delegation Link. Warden is a registered student for the Course.
Main Flow	1. Coordinator generates Warden Delegation Link via PWA. 2. Link sent to Warden. 3. Warden opens link; GPS geofence verified. 4. Warden's device becomes Temporary Edge Server. 5. Lecturer scans Gate-Open QR on Warden's screen. 6. Students scan personal QRs within proximity of Warden. 7. Session reaches scheduled end. 8. Warden presses 'Submit Session' (or auto-submit triggers at T+5 min after session end). 9. Encrypted data package delivered to Coordinator's account. 10. Coordinator reviews and approves via PWA.

Alternative Flows	A1: GPS geofence fails → link does not activate; DQA notified. A2: Warden does not submit → auto-submit at T+5 min post-session-end. A3: Coordinator does not approve within 48 hours → SYNC_OVERDUE alert to QA Officer.
Postconditions	Session data stored in Coordinator's Sync Queue with WARDEN_VERIFIED flag. VC Dashboard displays session with delegation indicator.

6.3 UC-03: DQA Officer Resets Student Device Binding

Field	Description
Use Case ID	UC-03
Name	Device Binding Reset
Actors	QA Officer (primary), Student (beneficiary)
Preconditions	Student's device binding count is below 2 for the current semester, or QA Officer provides an emergency override with DQA Director approval.
Main Flow	1. QA Officer logs into dashboard. 2. Searches for student by Registration Number. 3. Views hardware binding record. 4. Selects 'Reset Device Binding', enters reason code. 5. System clears the hardware_fingerprint field for that student. 6. System logs the reset event (QA Officer ID, timestamp, reason, old fingerprint hash). 7. Student's next QR scan will re-bind the new device. 8. Reset count incremented; if at limit, next reset requires DQA Director co-approval.
Postconditions	Student's hardware_fingerprint cleared. Reset logged for audit. Student able to scan with new device at next session.

7. Data Requirements

7.1 Core Entity Definitions

The QAAT Sidecar Schema comprises the following primary entities and their key attributes:

7.1.1 Students_Extended Table

Column Name	Data Type	Description
student_id	VARCHAR(50) PK	University Registration Number. Foreign key to SIS master record.
full_name	VARCHAR(200)	Student full name. Imported from SIS.
email	VARCHAR(255)	Official university email. Used for QR delivery.
course_id	VARCHAR(50) FK	Links to Courses table. Determines assigned Coordinator.
intake_session	VARCHAR(20)	e.g., Morning, Evening, Weekend.
current_year	SMALLINT	Year of study (1-6).
semester	SMALLINT	Current semester (1 or 2).
academic_year	VARCHAR(20)	e.g., 2025/2026.
enrollment_status	ENUM	ACTIVE, SUSPENDED, GRADUATED, WITHDRAWN.
hardware_fingerprint	VARCHAR(128)	AES-256 encrypted synthetic device hash.
rebind_count	SMALLINT	Number of device binding resets in current semester (max 2).
last_rebind_date	TIMESTAMP	Timestamp of last device binding reset.
qr_public_key_hash	VARCHAR(128)	Hash of the Public Key used to sign this student's QR.
qr_serial_number	VARCHAR(64)	Unique serial for current QR. Rotated on reissuance.
tenant_id	UUID FK	Tenant isolation key. RLS policy applied.

7.1.2 Sessions Table

Column Name	Data Type	Description
session_id	UUID PK	Globally unique session identifier.
coordinator_id	VARCHAR(50) FK	The Coordinator who initialised the session.
unit_id	VARCHAR(50) FK	The Course Unit for this session.

lecturer_id	VARCHAR(50) FK	Lecturer who performed Gate-Open scan.
venue_id	VARCHAR(50) FK	Assigned lecture hall/room.
session_date	DATE	Calendar date of the session.
gate_open_time	TIMESTAMP	When the Lecturer scanned Gate-Open.
gate_close_time	TIMESTAMP	When the Lecturer scanned Gate-Close (nullable).
checkin_window_start	TIMESTAMP	When student scanning was permitted.
checkin_window_end	TIMESTAMP	When student scanning was locked.
coordinator_end_time	TIMESTAMP	When Coordinator manually ended session (nullable).
auto_close_time	TIMESTAMP	When system auto-killed the session (nullable).
session_status	ENUM	PENDING_LECTURER, ACTIVE, CLOSED, AUTO_CLOSED.
warden_id	VARCHAR(50)	Registration Number of Warden if delegation used (nullable).
sync_status	ENUM	PENDING, SYNCED, FAILED.
audit_flags	TEXT[]	Array of audit flag codes (AUTO_CLOSED, WARDEN_VERIFIED, etc.).
tenant_id	UUID FK	Tenant isolation key.

7.1.3 Attendance_Logs Table

Column Name	Data Type	Description
log_id	UUID PK	Unique log entry identifier.
session_id	UUID FK	Links to Sessions table.
student_id	VARCHAR(50) FK	Student's Registration Number.
checkin_timestamp	TIMESTAMP	Exact time of successful attendance verification.
device_fingerprint_hash	VARCHAR(128)	Hash of device fingerprint used at time of scan.
beacon_rssi	SMALLINT	RSSI value (dBm) recorded at time of scan.
sequence_number	INTEGER	Order of this check-in within the session.
entry_method	ENUM	QR_SCAN, MANUAL_OVERRIDE.
override_officer_id	VARCHAR(50)	QA Officer ID if manual override (nullable).
override_reason	TEXT	Reason for manual override (nullable).

audit_flags	TEXT[]	e.g., DEVICE_MISMATCH_WARN, PROXIMITY_BORDERLINE.
tenant_id	UUID FK	Tenant isolation key.

7.2 Data Retention Policy

In accordance with academic data governance standards:

- Active semester attendance logs: retained in 'Hot' production database for immediate access.
- Completed semester logs: archived to compressed 'Warm' storage after semester examinations are graded. Accessible via dashboard but not in the active index.
- Historical logs older than 7 academic years: purged from the QAAT Sidecar Schema. Only aggregated attendance percentages are retained on the student's permanent transcript.
- Hardware fingerprint data: purged at the end of each academic year and re-captured on first scan of the new year.
- Admin audit logs: retained for a minimum of 7 years in compliance with institutional governance requirements.

8. Deployment and Operational Environment

8.1 Deployment Architecture

The QAAT cloud infrastructure is deployed using a containerised microservices architecture managed by Kubernetes on a commercial cloud provider. The primary service components are: the Authentication Service (JWT issuance and validation), the QR Generation Microservice (RSA key management and QR image generation), the Session Management Service (session lifecycle API), the Synchronisation Receiver Service (Chunked Resume-Sync endpoint), the Reporting Engine (data aggregation for dashboards), and the Notification Service (email delivery via SMTP relay).

The PostgreSQL database cluster is deployed in a primary-replica configuration with synchronous streaming replication for zero data loss failover. A Redis cluster provides session caching and synchronisation queue management. A CDN with edge caching serves the Daily Manifest JSON files to Coordinator PWAs globally, ensuring sub-5-second fetch times regardless of geographic location.

8.2 Physical Hardware Deployment (BLE Beacons)

BLE Beacons must be physically installed in each lecture venue prior to QAAT deployment. Each Beacon must be mounted at a height of 2–3 metres, preferably at the front of the room near the Lecturer's station, to ensure maximum proximity accuracy and minimum wall-reflection interference. Each Beacon must be registered in the QAAT Venue Registry with its UUID, associated room identifier, and physical coordinates. Universities are responsible for beacon procurement, installation, power management, and firmware maintenance. Recommended Beacon specification: iBeacon-compatible, BLE 5.0, minimum 2-year battery life on CR2477 or equivalent.

8.3 Rollout Phases

Phase	Duration	Activities
1	Weeks 1–2	Tenant onboarding; DQA configuration; student data import from SIS; QR code generation and email delivery; Coordinator PWA distribution.
2	Weeks 3–4	Beacon procurement, installation, and UUID registration; RSSI threshold calibration per venue; Coordinator and Warden training.
3	Weeks 5–6	Controlled pilot deployment in one department; monitoring of sync pipeline; fraud detection calibration; feedback collection.
4	Weeks 7–8	Full campus rollout; VC and DQA dashboard activation; live monitoring; post-deployment audit.

Appendix A: CSV Import Format Specification

The following CSV format is used for manual student data import into QAAT when SIS API access is unavailable. All fields are mandatory unless marked optional.

Column Header	Type	Required	Notes
student_id	String	Yes	University Registration Number. Max 50 chars. Unique key.
full_name	String	Yes	Student full name. Max 200 chars.
email	String	Yes	Official university email. Must be valid RFC 5322 format.
course_id	String	Yes	Course code (e.g., BIT, BCom). Must match Courses table.
intake_session	String	Yes	Values: Morning, Evening, Weekend, Distance.
current_year	Integer	Yes	Year of study. Values: 1 through 6.
semester	Integer	Yes	Values: 1 or 2.
academic_year	String	Yes	Format: YYYY/YYYY (e.g., 2025/2026).
enrollment_status	String	Yes	Values: ACTIVE, SUSPENDED, GRADUATED, WITHDRAWN.

Appendix B: Glossary of Audit Flag Codes

Flag Code	Meaning and Action
AUTO_CLOSED	Session terminated by 3-hour automatic expiry. Coordinator did not manually end session. QA Officer review recommended.
WARDEN_VERIFIED	Session was hosted by a delegated Warden. Coordinator approval obtained. Acceptable but flagged for VC transparency.
DEVICE_MISMATCH	Student scan rejected: device fingerprint did not match registered binding. Possible proxy attempt. QA Officer review required.
DUPLICATE_SCAN_ATTEMPT	Same student attempted to scan more than once in the same session. Second scan silently rejected. Logged for anomaly analysis.
INCOMPLETE_LECTURER_LOG	Lecturer Gate-Close scan was not performed. Session end time is auto-recorded. QA Officer review recommended.
SYNC_OVERDUE	Session data has not been synchronised within 48 hours of session end. QA Officer should contact Coordinator.

MANUAL_OVERRIDE	Attendance record was added manually by a QA Officer, not via student scan. Full audit trail recorded.
PROXIMITY_BORDERLINE	Student check-in accepted but RSSI was within 5 dBm of the lower threshold boundary. Not an error; informational flag.
GHOST_LECTURE_SUSPECTED	Lecturer Gate-Open recorded but student attendance below 5% of enrolled class. DQA Director review required.
GPS_GEOFENCE_FAIL	Warden Delegation Link activation rejected because device GPS was outside the registered venue boundary.

Appendix C: IEEE / ISO Standards Compliance Summary

Standard	Clause / Section	SRS Coverage
IEEE Std 830-1998	Section 3 — Specific Requirements	Section 3 of this document.
IEEE Std 830-1998	Section 4 — Introduction Structure	Section 1 of this document.
IEEE Std 830-1998	Section 5 — Specific Req. Attributes	NFR sections 3.2.1–3.2.6.
ISO/IEC/IEEE 29148:2018	5.2 — Stakeholder Requirements	Section 2.4 — User Classes.
ISO/IEC/IEEE 29148:2018	5.3 — System Requirements	Sections 3.1 and 3.2.
ISO/IEC/IEEE 29148:2018	5.4 — Software Requirements	Section 3.1 FR-01 through FR-10.
ISO/IEC 25010:2011	Quality Model — Security	Sections 3.2.3 and 5.1.
ISO/IEC 25010:2011	Quality Model — Reliability	Section 3.2.2.
ISO/IEC 25010:2011	Quality Model — Usability	Section 3.2.4.
ISO/IEC 25010:2011	Quality Model — Maintainability	Section 3.2.5.

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